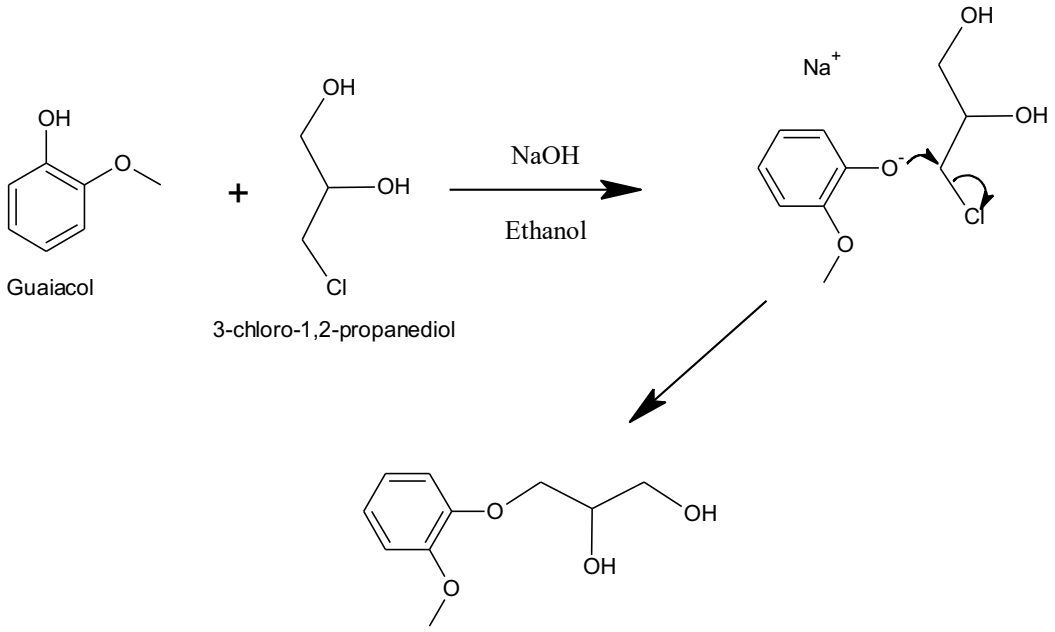
The Williamson ether synthesis is named for its inventor, Alexander Williamson who discovered the reaction in 1850. This reaction, since its discovery, is an important method of producing a vast array of aromatic and alkyl ethers. The reaction is essentially an S_N2 reaction between an alkoxide or a phenoxide with an alkyl halide. The alkoxides can be formed by the reaction of the corresponding alcohols with NaH, likewise, the phenoxides can be formed from the corresponding phenols with base such as NaOH, both compounds are good nucleophiles in the subsequent nucleophilic substitution.

Guaifenesin is an expectorant, used to relieve congestion from mucus. Familiar over-the-counter medications that contain the compound are Mucinex and Robitussin. Guaifenesin is an ether that is made by the S_N2 reaction of the phenoxide of guaiacol and 3-chloro-1,2-propanediol by the Williamson synthesis.



Guaifenesin

Safety Precautions:

- **3-Chloro-1,2-propanediol**: Highly toxic by skin absorption, ingestion, and inhalation, targeting the central nervous system. It is also a suspected carcinogen. Proper precautions such as wearing protective gloves, goggles, and a lab coat, as well as working in a hood, are necessary when handling this chemical.
- **Guaiacol:** Toxic and should be handled with caution. Specific precautions should be taken to minimize exposure.
- **Sodium hydroxide:** Toxic and causes severe burns. Protective measures, including appropriate personal protective equipment and handling procedures, should be followed.
- **Aqueous ethanol, ethyl acetate, and hexanes:** Flammable solvents. Precautions to prevent ignition and ensure safe handling should be taken.

Williamson Ether Synthesis (Micro Scale)

In a fume hood:

1. In a 25mL RB (round bottom) flask add 0.55mL guaiacol in 3 mL of EtOH (Ethanol)
2. Add 1 mL of 6M NaOH solution dropwise.
3. Fit the flask with a reflux condenser and reflux for 10 min in a sand bath
4. Add 1.0 mL 3-chloro-1,2-propanediol solution (50% by volume in ethanol solution) dropwise over 10 min.
5. Reflux for 1 hour
6. After an hour of reflux cool to room temperature, remove stir bar.
7. Transfer solution to a WEIGHED vial with a pipette and cover with a kimwipe and rubber band.
8. Store the vial until next week.

The following week:

1. After a week, the product should be dry so
2. Wash solid/oil with 2 x 1 mL EtOAc (Ethyl Acetate)
3. Put the EtOAc washes in a separate WEIGHED vial
4. Then run a TLC of your product versus pure guaifenesin (50:50 hexane/EtOAc)
5. Remove the EtOAc then weigh the vial.

If the contents of the vial are not dry then…

1. Next week add 1 mL of water to dissolve the salts
2. Extract with 2 x 3 mL of EtOAc (Ethyl Acetate)
3. Dry with $MgSO_4$ and continue with the experiment as described above.

Results and Conclusion:

An NMR of the starting materials and the final product will be given to you/run by you.

You must interpret the NMRs by drawing the product on the spectra and putting arrows from the protons on the structure to the resonances on the spectra.

Calculate the percent yield. Interpret your results.

Calculate the Rf from the TLC. Interpret your results.